

205 q/zq :

$\mathbb{N} \xleftarrow{(\mathbb{N}, +, \cdot, <)} \text{existence}$

given by Peano's Axioms $N1 - N5$

Hammer This

$$\mathbb{Z} = \mathbb{N} \times \mathbb{N} / \sim = \left\{ [m, n] \mid m, n \in \mathbb{N} \right\}$$

we say $(m, n) \sim (p, q)$ if $m + q = n + p$

• operations: $+_{\mathbb{Z}}, \cdot_{\mathbb{Z}}, <$

• $(\mathbb{N}, +, \cdot, <) \subset (\mathbb{Z}, +_{\mathbb{Z}}, \cdot_{\mathbb{Z}}, <) \subset (\mathbb{Q}, +_{\mathbb{Q}}, \cdot_{\mathbb{Q}}, <)$

$$\mathbb{Q} = \left\{ \begin{bmatrix} b & a \\ a & 0 \end{bmatrix} : a, b \in \mathbb{Z}, a \neq 0 \right\}$$

↖ Motivation: Include

$$ax = b, \quad a, b \in \mathbb{Z} \\ \& \text{ don't want } a=0$$

Something New:

$$[x, y] +_{\mathbb{Q}} [0, 1] = [x, y]$$

$$[x, y] \cdot_{\mathbb{Q}} [1, 1] = [x, y]$$

$$[x, y] \cdot_{\mathbb{Q}} [y, x] = [1, 1]$$

Division Operation:

if $[z, w] \in \mathbb{Q}$ with $z, w \neq 0$, we have its inverse

$$[w, z] := [z, w]^{-1} \text{ or } \frac{1}{[z, w]}$$

Then for any $[x, y] \in \mathbb{Q}$ we can define the division operation

$$\frac{[x, y]}{[z, w]} := [x, y] \cdot [z, w]^{-1}$$

Prop: $(\mathbb{Q}, +, \cdot)$ is a field

Recall:

Defⁿ: Let F be a set, F is called a field if there are two operations $+$, $\cdot$ on F satisfying the following 11 properties

- $\mathbb{N}$ A1) $\forall a, b \in F \Rightarrow a+b \in F$ $\mathbb{N}$ M1) $\forall a, b \in F \Rightarrow a \cdot b \in F$
- $\mathbb{N}$ A2) $a+b = b+a$ $\mathbb{N}$ M2) $a \cdot b = b \cdot a$
- $\mathbb{N}$ A3) $a+0 = a$ $\mathbb{Z}$ M3) $\exists 1 \in F \mid a \cdot 1 = a$
- $\mathbb{N}$ A4) $a+(b+c) = (a+b)+c$ $\mathbb{Z}$ M4) $a \cdot (b \cdot c) = (a \cdot b) \cdot c$
- $\mathbb{Z}$ A5) $\forall a \in F, \exists -a \in F \mid a+(-a)=0$ $\mathbb{Q}$ M5) $\forall a \in F, a \neq 0 \exists a^{-1} \in F \mid a \cdot a^{-1} = 1$
- $\mathbb{N}$ DL) $a \cdot (b+c) = (a \cdot b) + (a \cdot c) = a \cdot b + a \cdot c$

Indeed: we can also define order structure on $\mathbb{Q}$

$(\mathbb{Z}, <_{\mathbb{Z}})$ $\forall [m, n], [p, q] \in \mathbb{Q}$ we say

$[m, n] <_{\mathbb{Q}} [p, q]$ if $m \cdot q <_{\mathbb{Z}} p \cdot n$

Checks: $<_{\mathbb{Q}}$ is order $(0_1, 0_2)$

Def: Let F be a set " L " is an order structure on F

O1) $\forall a, b \in F$ one of the following holds
 $a < b$, $b < a$, $a = b$

O2) $a < b$ & $b < c \Rightarrow a < c$

Defⁿ: (ordered field). Let $(F, +, \cdot)$ be a field, " L " is an order structure on F . We say $(F, +, \cdot, L)$ is an ordered field if we have

O3) (" L " is compatible w/ " $+$ ") $a < b \Rightarrow a+c < b+c \quad \forall c$

O4) (" L " is compatible w/ " $\cdot$ ") $a < b, c > 0 \Rightarrow a \cdot c < b \cdot c$

Prop: $(\mathbb{Q}, +, \cdot, <)$ is an ordered field.

Problem with $\mathbb{Q}$: $\exists$ "holes" on $\mathbb{Q}$.

Eg: Prove $\sqrt{2} \notin \mathbb{Q}$ (omitted)

$$\mathbb{N} \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow ?$$

Get a larger set $\mathbb{Q} \rightarrow \mathbb{R}$??

Keep in mind:

- include $\mathbb{Q}$ as a proper subset
- hope $\mathbb{R}$ has ordered field structure

Two different constructions:

- Dedekind (uses "cut")
- Cantor (uses "Cauchy sequence")

Dedekind:

Idea: $\mathbb{R}$ = a collection of "special" subsets of $\mathbb{Q}$

Def: A subset α of $\mathbb{Q}$ is called a cut if it satisfies

(I) $\alpha \neq \emptyset, \alpha \neq \mathbb{Q}$ (α proper subset of $\mathbb{Q}$)

(II) if $p \in \alpha, q \in \mathbb{Q}, q < p \Rightarrow q \in \alpha$
(α is closed downward)

(III) if $p \in \alpha$, then $\exists r \in \alpha \mid p < r$
(α has no greatest element)

Define:

$$\mathbb{R} = \{ \alpha : \alpha \text{ is a cut in } \mathbb{Q} \}$$

Each element is a cut

• $\mathbb{Q} \subseteq \mathbb{R}$:

Take $r \in \mathbb{Q}$, consider r^+ to

$$r^+ = \{ p \in \mathbb{Q} : p < r \} \subset \mathbb{Q}$$

$\rightarrow r^+$ is a cut $\Rightarrow r^+ \in \mathbb{R}$

$$\boxed{\begin{array}{ccc} \mathbb{Q} & \longrightarrow & \mathbb{R} \\ r & \longmapsto & r^+ \end{array}}$$

Eg:

$$\alpha = \{ p \in \mathbb{Q} : p < 1 \} \rightsquigarrow \text{cut.}$$

$$(III) p_0 \in \alpha \Rightarrow p_0 < 1$$

$$r = p_0 + \frac{1 - p_0}{2}$$

Not a cut

$$\cancel{\alpha} = \{ p \in \mathbb{Q} : p \leq 1 \}$$

• Order structure on $\mathbb{R}$: for any $\alpha, \beta \in \mathbb{R}$, we say

$$\alpha < \beta \text{ if } \alpha \subsetneq \beta$$

Takeaway: we can view $\mathbb{Q}$ as a subset of $\mathbb{R}$.

Structure: $\alpha, \beta \in \mathbb{R}$

$+_{\mathbb{R}}$:

$$\alpha +_{\mathbb{R}} \beta := \left\{ a + b : a \in \alpha, b \in \beta \right\}$$

Check: $\alpha + \beta$ is a cut.

(II) $p \in \alpha + \beta \Rightarrow p = a + b$, for some $a \in \alpha, b \in \beta$

goal: if $q \in \mathbb{Q} \mid q < p \Rightarrow q = \boxed{} + \boxed{}$

$$q = \underbrace{\left(a - \frac{p-q}{2} \right)}_{< a} + \underbrace{\left(b - \frac{p-q}{2} \right)}_{< b}$$

$\leftarrow \text{since } q < p \rightarrow$

(III) $p \in \alpha + \beta \Rightarrow p = a + b$ for some $a \in \alpha, b \in \beta$

goal: $\gamma \in \alpha + \beta$ s.t. $p < \gamma$

$$\left. \begin{array}{l} a \in \alpha \Rightarrow a' \in \alpha, a < a' \\ b \in \beta \Rightarrow b' \in \beta, b < b' \end{array} \right\} r = a' + b' \in \alpha + \beta$$

• $\exists 0 \mid a + 0 = a \quad \forall a \in \mathbb{R}$

$$\text{In } \mathbb{R}: 0^* = \left\{ p \in \mathbb{Q} : p \leq 0 \right\}$$

$$\forall \alpha \in \mathbb{R}, \alpha + 0^* = \left\{ a + b : a \in \alpha, b \in 0^* \right\} \neq \alpha \quad (\leq \geq)$$

• A5 In $\mathbb{R}$: $\alpha \in \mathbb{R}$

Goal: introduce $-\alpha \mid -\alpha \in \mathbb{R} \text{ \& } \alpha + (-\alpha) = 0^*$

Eg:



$$\alpha = \{ p \in \mathbb{Q} \mid p < 3 \}$$

(for inverse of α :
intuitively, take the
reflection

So, $-\alpha := \{ p \in \mathbb{Q} : p < -a \text{ for some } a \in \alpha \}$

Multiplication:

$$\alpha \cdot \beta := \{ p \in \mathbb{Q} : p < a \cdot b \quad a \in \alpha, b \in \beta \}$$

$$\alpha \cdot \beta := \begin{cases} -(1-\alpha) \cdot \beta & \text{if } \alpha < 0^*, \beta > 0^* \\ \alpha > 0^*, \beta < 0^* \\ \alpha < 0^*, \beta < 0^* \end{cases}$$

Proposition: $(\mathbb{R}, +_{\mathbb{R}}, \cdot_{\mathbb{R}}, <_{\mathbb{R}})$ is an ordered field!

which includes $(\mathbb{Q}, +_{\mathbb{Q}}, \cdot_{\mathbb{Q}}, <_{\mathbb{Q}})$ as a subfield

Next Morality:

$\mathbb{R}$ has

least upper bound property

→ completeness
Axiom